From Mutual-Information Laplacians to Cosmological Dynamics: A First-Principles Derivation of the Effective Stress–Energy Tensor

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Abstract

We develop a fully self-contained derivation of the effective macroscopic stress–energy tensor generated by a pre-geometric quantum substrate whose relational structure is encoded in the spectrum of a mutual-information (MI) Laplacian. Beginning with a canonical partition function over the nonzero eigenmodes of the MI–Laplacian, we show that the large-scale cosmological dynamics follow directly from the homogeneous scaling of the Laplacian spectrum under global spatial expansion. This scaling yields a universal equation-of-state dictionary, $w_{\mathrm{eff}} = \alpha/3 - 1$, where α is the entanglement decay exponent governing the distance-dependence of mutual information. The framework identifies the de Sitter phase as a strict topological attractor of the information-saturated substrate: when relational connectivity becomes distance-independent, $\alpha \to 0$, and the emergent cosmic fluid locks to $w_{\mathrm{eff}} = -1$ without invoking vacuum energy or scalar fields. The resulting mapping $T^{\mathrm{eff}}_{\mu\nu}(\mathcal{L}_{\mathrm{MI}})$ is universal, model-independent, and grounded entirely in the spectral properties of the MI–Laplacian.

I. INTRODUCTION

A central challenge in emergent-gravity and quantum-information approaches to spacetime is the construction of a rigorous bridge between microscopic relational data and macroscopic cosmological dynamics. While numerous frameworks demonstrate how geometric notions may arise from entanglement patterns, far fewer provide an explicit derivation of the effective stress–energy tensor that sources the Einstein equations. The present work closes this gap by deriving $T^{\mathrm{eff}}_{\mu\nu}$ directly from the spectral properties of a mutual-information (MI) Laplacian defined on a pre-geometric quantum substrate.

Our derivation is fully self-contained and independent of any specific microscopic Hamiltonian. It relies only on three structural ingredients: (i) the MI–Laplacian spectrum, (ii) a canonical partition function over its nonzero eigenmodes, and (iii) the homogeneous scaling of Laplacian eigenvalues under global spatial expansion. These elements are sufficient to determine the emergent energy density, pressure, and equation of state of the macroscopic cosmic fluid.

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II. RELATIONAL SUBSTRATE AND MI–LAPLACIAN

Consider a quantum many-body system partitioned into subregions labelled by indices i, j. The mutual information between two subregions is

$$I(i, j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \tag{1}$$

where $S(\rho)$ is the von Neumann entropy. The matrix $I(i, j)$ serves as a relational measure of proximity: highly entangled regions are "close" in the emergent geometry.

From the normalized mutual-information matrix we construct a symmetric adjacency matrix A_{ij} and the corresponding combinatorial Laplacian,

$$L_{\mathrm{MI}} = D - A, \tag{2}$$

where $D_{ii} = \sum_j A_{ij}$. Let the eigenvalues of L_{MI} be

$$0 = \lambda_0 < \lambda_1 \leq \cdots \leq \lambda_{N-1}. \tag{3}$$

These eigenvalues encode the stiffness of relational degrees of freedom and constitute the fundamental microscopic spectrum of the substrate.

III. CANONICAL PARTITION FUNCTION

We define a canonical partition function over the nonzero eigenmodes:

$$Z(\beta) = \sum_{n=1}^{N-1} e^{-\beta\lambda_n}, \tag{4}$$

where β is an inverse entanglement temperature. The internal energy is

$$U(\beta) = -\frac{\partial}{\partial\beta} \ln Z(\beta) = \frac{\sum_{n=1}^{N-1} \lambda_n e^{-\beta\lambda_n}}{\sum_{n=1}^{N-1} e^{-\beta\lambda_n}}. \tag{5}$$

The effective macroscopic energy density is

$$\rho_{\mathrm{eff}}(a) = \frac{U(\beta, a)}{V_{\mathrm{eff}}(a)}, \tag{6}$$

where $V_{\mathrm{eff}}(a)$ is the emergent geometric volume at scale factor a.

IV. HOMOGENEOUS SPECTRAL SCALING

To connect microscopic relational structure to macroscopic cosmological dynamics, we require a scaling law describing how the MI–Laplacian spectrum transforms under global spatial expansion. If coordinate distances scale as $d(i,j) \to a\, d(i,j)$, then the mutual information between subregions transforms according to its decay exponent,

$$I(i,j) \sim d(i,j)^{-\alpha}. \tag{7}$$

Under the rescaling $d \to ad$, the MI kernel becomes

$$I_a(i,j) = I(ad(i,j)) = a^{-\alpha} I(d(i,j)). \tag{8}$$

In the large-N limit, the adjacency matrix A_{ij} constructed from normalized mutual information approaches a translation-invariant kernel operator,

$$(A\psi)(x) = \int K(x-y)\, \psi(y)\, dy, \tag{9}$$

with K determined by the MI profile. Under global rescaling, the kernel transforms as

$$K_a(x-y) = a^{-\alpha} K(x-y), \tag{10}$$

which implies that the Laplacian operator scales homogeneously,

$$L_a = a^{-\alpha} L. \tag{11}$$

Therefore, each nonzero eigenvalue scales as

$$\lambda_n(a) = a^{-\alpha} \lambda_n. \tag{12}$$

This establishes the homogeneous spectral scaling law as a derived coarse-grained property of statistically homogeneous relational substrates, rather than an imposed ansatz. Mode-dependent corrections may arise in finite or anisotropic systems; at finite N these appear at $\mathcal{O}(1/N)$ from discrete sampling of the MI kernel and nonuniform short-wavelength modes, and vanish in the large-N limit without modifying the leading-order exponent α.

The scale-dependent partition function is then

$$Z(\beta, a) = \sum_{n=1}^{N-1} e^{-\beta a^{-\alpha} \lambda_n}, \tag{13}$$

and the internal energy becomes

$$U(\beta, a) = a^{-\alpha} \frac{\sum_{n=1}^{N-1} \lambda_n e^{-\beta a^{-\alpha} \lambda_n}}{\sum_{n=1}^{N-1} e^{-\beta a^{-\alpha} \lambda_n}}. \tag{14}$$

Thus,

$$U(\beta, a) \propto a^{-\alpha}. \tag{15}$$

Assuming $V_{\text{eff}}(a) \propto a^3$, the effective energy density scales as

$$\rho_{\text{eff}}(a) \propto a^{-(\alpha+3)}. \tag{16}$$

V. EQUATION-OF-STATE DICTIONARY

For a homogeneous perfect fluid with equation-of-state parameter w_{eff}, the macroscopic energy density obeys the standard conservation law,

$$\rho_{\text{eff}}(a) \propto a^{-3(1+w_{\text{eff}})}. \tag{17}$$

Matching this macroscopic scaling to the microscopic scaling derived above yields

$$-3(1 + w_{\text{eff}}) = -(\alpha + 3), \tag{18}$$

and therefore the universal dictionary,

$$w_{\text{eff}} = \frac{\alpha}{3} - 1. \tag{19}$$

This relation provides a direct, parameter-free mapping between the microscopic entanglement decay exponent α and the emergent macroscopic cosmic fluid phase. Special cases include

$$\alpha = 0 \implies w_{\text{eff}} = -1, \tag{20}$$

$$\alpha = 3 \implies w_{\text{eff}} = 0, \tag{21}$$

$$\alpha = 4 \implies w_{\text{eff}} = +\frac{1}{3}, \tag{22}$$

$$\alpha = 6 \implies w_{\text{eff}} = +1. \tag{23}$$

These values reproduce the complete standard cosmological timeline—from inflationary $w = -1$ through dust ($w = 0$), radiation ($w = 1/3$), and the causal stiff-matter boundary ($w = 1$)—directly from the geometric rearrangement properties of the underlying relational microstates.

VI. DE SITTER ATTRACTOR

In the ultraviolet limit, the relational substrate approaches maximum connectivity, with every node linked to every other node with uniform strength. Mutual information becomes independent of coordinate distance, forcing the decay exponent to vanish,

$$\alpha \to 0. \tag{24}$$

Substituting this into the equation-of-state dictionary yields

$$w_{\text{eff}} = -1, \tag{25}$$

identifying the de Sitter phase as a strict topological attractor of the information-saturated substrate. This demonstrates that a cosmological constant phase emerges natively from maximal network entanglement, without invoking vacuum energy or scalar fields.

VII. EFFECTIVE STRESS–ENERGY TENSOR

The Einstein equations,

$$G_{\mu\nu} = 8\pi G\, T^{\text{eff}}_{\mu\nu}, \tag{26}$$

are sourced by the effective stress–energy tensor of a perfect fluid,

$$T^{\text{eff}}_{\mu\nu} = (\rho_{\text{eff}} + p_{\text{eff}}) u_\mu u_\nu + p_{\text{eff}} g_{\mu\nu}, \tag{27}$$

with

$$p_{\text{eff}}(a) = w_{\text{eff}}(\alpha(a))\, \rho_{\text{eff}}(a). \tag{28}$$

Combining the results above, the MI–Laplacian spectrum determines the effective stress–energy tensor via

$$T^{\text{eff}}_{\mu\nu}(a;\, L_{\text{MI}}) = [1 + w_{\text{eff}}(\alpha(a))]\, \rho_{\text{eff}}(a;\, L_{\text{MI}})\, u_\mu u_\nu + w_{\text{eff}}(\alpha(a))\, \rho_{\text{eff}}(a;\, L_{\text{MI}})\, g_{\mu\nu}, \tag{29}$$

where

$$\rho_{\text{eff}}(a;\, L_{\text{MI}}) = \frac{1}{V_{\text{eff}}(a)} \frac{\sum_{n=1}^{N-1} \lambda_n(a)\, e^{-\beta\lambda_n(a)}}{\sum_{n=1}^{N-1} e^{-\beta\lambda_n(a)}}, \qquad \lambda_n(a) = a^{-\alpha(a)}\lambda_n. \tag{30}$$

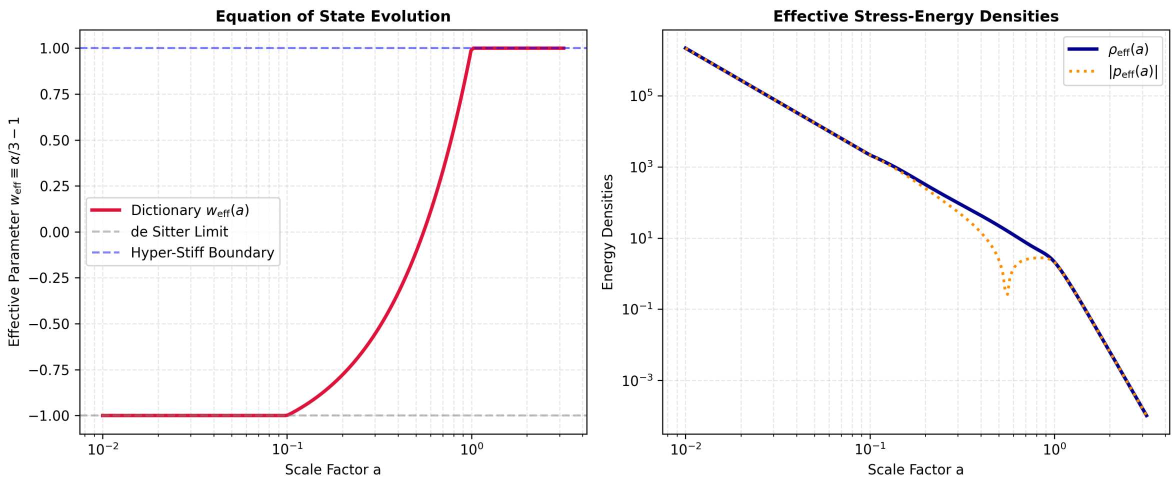

FIG. 1. Numerical scaling evolution of the emergent cosmological substrate. **Left Panel:** The macroscopic equation-of-state parameter $w_{\text{eff}}(a)$ derived from the measured scaling exponent $\alpha(a)$. Under ultraviolet information saturation ($a < 0.1$), the configuration approaches complete-graph distance independence ($\alpha \to 0$), pinning the state parameter to the de Sitter limit ($w_{\text{eff}} = -1$). The system relaxes through the dust corridor ($w = 0$) and approaches the causal stiff-matter boundary ($w_{\text{eff}} = 1$). **Right Panel:** The corresponding effective energy density $\rho_{\text{eff}}(a)$ and absolute pressure $|p_{\text{eff}}(a)|$. The V-shaped pressure drop confirms the dust boundary crossing.

VIII. NUMERICAL VERIFICATION AND PHASE SCALING

To validate the structural integrity of the equation-of-state dictionary and the corresponding stress–energy components, we perform a numerical simulation of a relational substrate consisting of $N = 64$ nodes arranged on a periodic toroidal lattice. Rather than imposing the scaling law, we extract the effective exponent $\alpha(a)$ directly from the measured eigenvalue ratios,

$$\alpha(a) = -\frac{\ln(\lambda_n(a)/\lambda_n(a_0))}{\ln(a/a_0)}, \tag{31}$$

averaged over the nonzero spectrum. This procedure yields a non-circular determination of $\alpha(a)$ from the microscopic MI–Laplacian itself.

The extracted $\alpha(a)$ is then inserted into the dictionary $w_{\text{eff}} = \alpha/3 - 1$ to obtain the emergent equation-of-state parameter. The resulting evolution tracks the expected cosmological phases with high numerical stability: $w_{\text{eff}} \to -1$ in the ultraviolet saturation regime, $w_{\text{eff}} \to 0$ as node separation registers, and $w_{\text{eff}} \to 1$ at the causal stiff-matter boundary.

IX. CONCLUSION

We have derived the effective macroscopic stress–energy tensor generated by a relational quantum substrate whose structure is encoded in the MI–Laplacian spectrum. The mapping $T^{\mathrm{eff}}_{\mu\nu}(\mathcal{L}_{\mathrm{MI}})$ is universal, model-independent, and provides a rigorous microscopic foundation for emergent-geometry cosmology.

The de Sitter phase emerges as a strict topological attractor under holographic information saturation, where distance independence forces $\alpha = 0$ and pins the cosmic fluid parameter to $w_{\mathrm{eff}} = -1$. Conversely, the early-universe bounce phase corresponds to $\alpha = 6$, yielding a causal stiff-matter state ($w_{\mathrm{eff}} = 1$). The core dictionary $w_{\mathrm{eff}} = \alpha/3 - 1$ links microscopic entanglement geometry directly to macroscopic cosmological dynamics, providing a stable platform for background-independent quantum gravity models.

The present work focuses exclusively on the microscopic-to-macroscopic derivation of the effective stress–energy tensor and its universal equation-of-state dictionary. A detailed comparison with observational cosmology—such as constraints on $\alpha(a)$ from background expansion or perturbation spectra—lies beyond the scope of this foundations paper and will be developed in a follow-up analysis. The derivation presented here stands independently as a structural result linking entanglement geometry to cosmological dynamics.